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SCHOOL OF COMPUTER SCIENCE

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END:	**13ND MARCH 2025, FRIDAY**	

JANURARY 2025 – SEMESTER 10

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- ✔ **I recognize that should evidence emerge that my work fails to comply with either of the above declarations, then I may be liable to proceeding under regulation.**

Title:	**"Data-Driven Insights through Interactive Visualization for Real-World Decision Making"**

MLO2 – INDIVIDUAL ASSIGNMENT (30%)

Student Full Name	University ID	Student ID	Signature
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Title: Data-Driven Insights through Interactive Visualization for Real-World Decision Making

Task 1: Problem Definition & Context

1.1 Problem Statement

Road traffic accidents stand as one of the main public safety issues in Nepal. They are a preventable cause of the death and disability, yet road accidents continue to be the cause of fatalities and the country is already facing economic difficulties due to the losses caused by these accidents. The road network of Nepal is growing rapidly, which is also one of the reasons behind these accidents, but the main reasons are the lack of strictness in the implementation of the traffic rules and the geographical discontinuity which ranges from the Himalayan high-altitude routes to the crowded city areas. In fact, accident patterns are so varied that they differ from one province to another, different types of roads, and different time periods. This paper tries to address the main question of the study: Which factors lead to the highest number of road accidents and casualties in Nepal, and how can the use of data lead to the proper, evidence-based safety intervention across the seven provinces of the country? The question in itself is not descriptive: it is not aimed at counting the accidents but at discovering the cause and the related patterns between variables such as weather conditions, types of the road surface, vehicle categories, time of the day, driver demographics, and speed zones. These findings have a direct impact on deciding where to increase the traffic monitoring, how to distribute the funds for road safety infrastructure, and which groups of the drivers to target for public awareness campaigns.

1.2 Stakeholders

Those stakeholder groups mentioned here have direct interest in the knowledge this dashboard will provide:

- Traffic police and enforcers with the ability to utilize time and speed-related data to maximize patrol presence.
- Provincial and national transport planning authorities that deal with the investment of road infrastructure and safety control.
- Nepal Department of roads that have the capacity of giving priority to road maintenance in high risk districts.

- The Ministry of Health and Population policymakers in charge of responding to injury and fatality patterns.
- Insurance companies and vehicle regulatory bodies that use such factors as vehicle type and driver profile to perform their risk assessment
- Researchers and non-governmental organizations that conduct road safety and spread awareness campaigns.

1.3 Why Visualization is Needed

Road accident datasets usually contain multiple types of data, for example time-related, location-related, categorical and numeric variables all together. On-the-spot presentation of these data in traditional tables or charts is usually not informative enough to understand the interactions between different levels/dimensions of a variable. For instance, from the summary statistics alone one cannot tell if the night-time accidents leading to fatal injuries are more likely to happen on highways or that the relationship between rain and accidents is different in different provinces. Interactive visualization allows stakeholders with different levels of data analysis expertise to filter, compare and explore the data for themselves and thus find the insights that are most relevant to their specific decision-making situations. According to research done by Tufte (2001) and Cairo (2016), well-made visualizations not only help to reduce mental effort but also bring to light previously hidden patterns, and enhance the speed and correctness of data-driven decisions. In the area of public safety, where a delay or an improper targeting of an intervention can be fatal, this benefit is very important.

1.4 Decision-Making Objectives

The dashboard will respond to the five decision-relevant questions as indicated below:

- In which provinces or districts do accidents predominate, and are there some provinces or districts incomparably affected?
- Is there an increase or decrease in the peak of accidents and does it vary in severity or year-on-year direction?
- What are the most environmental and infrastructural factors (weather, road surface, road type) that are most related to fatal outcomes?
- What are the most common trying vehicle types and driver profiles that have the highest severity of accidents?

- What has been the year on year change in the accident trends and are interventions bringing about any observable change?

Task 2: Data Preparation & Understanding

2.1 Data Source

The data that will be used in this assignment is a simulated account on road traffic accidents in Nepal between January 2023 and January 2026. It was created artificially to be used in academic and data visualization learning and is not reflective of actual accident data. The dataset contains 1,850 records and 17 attributes, and it is a comfortable number to meet the 500-record and 5-attributes minimum requirements stated in the assignment brief. It has been generated by the means of Artificial Intelligence Tools such as ChatGPT that is meant to recreate the realistic distribution of accidents across the provinces and districts of Nepal based on the road network geography of this country.

Link: https://www.kaggle.com/datasets/monishshrestha/road-traffic-accident-dataset-nepal-2023-present

2.2 Dataset Structure and type of attributes.

Attribute	Type	Description
Accident_ID	Nominal	Unique record identifier.
Accident Date / Year/ Months	Temporal	Date fields for time-series analysis
Province / District	Spatial / Categorical	7 provinces and districts in Nepal.
Road/Intersection type	Categorical	Highway, Urban, Rural, Feeder; Junction, Curve, Straight.
Time of Day	Categorical	Morning, Afternoon, evening, night, day.
Accident_Severity	Ordinal	Fatal > Serious Injury > Minor Injury > Damage Only
Weather_Condition / Road_surface	Categorical	5 weather conditions; 4 road conditions.
Vehicle_Type	Categorical	There are 7 types of vehicles (Motorbike to Truck)
Speed_Zone	Numerical (Discrete)	Posted speed zone: 30–100 km/h
Driver_Gender	Categorical	Male, Female, Other

Attribute	Type	Description
Number_of_Vehicles / Casualties	Numerical	Count fields; Casualties: mean 5.08, max 25

Table 1: Dataset Attribute Reference

2.3 Data Cleaning & Preparation

- **Province–District Mapping Correction:** Systematic Province-District mis assignment in the entire dataset was the most severe data quality problem. As an example, Kalikot (Karnali Province) was commonly referred to Bagmati on records; and Kailali and Achham were referred to wrong provinces. To assign every district to the correct administrative province according to the official boundaries of Nepal, a Tableau Calculated Field called Correct Province was constructed using a nested IF/ELSEIF statement that links each district to its appropriate administrative province (Government of Nepal, 2015). This field substituted the raw Province column of every visualization.
- **Vehicle Type Grouping:** The raw Vehicle_Type field had seven different categories. To enhance the visual display of the dashboard in the vehicle comparison section, a calculated field, Grouped Vehicle Type was developed to group these into four major categories which included: Bus, Car (including Jeep and Microbus), Motorbike (including Scooter) and Truck. The grouping provides less visual clutter and still maintains the analytically significant differences between motorized two-wheelers, public transport, personal cars, and heavy goods vehicles.
- **Temporal Calculated Fields & Parameters:** Current Year Accidents/Casualties and Previous Year Accidents/Casualties were retrieved on the basis of dynamic parameters (Current Year, Previous Year). The directional KPI indicators were powered by computing Year-over-Year change: (CY Casualties - PY Casualties)/PY Casualties. Two other parameters Select Accident Severity and Select Province are used to apply dashboard-wide filters using calculated fields: [Select Accident Severity] = [Accident Severity] OR [Select Accident Severity] = All and the same thing under Select Province using Correct Province.
- **Severity Ordering & Color Grammar:** Accident Severity is an ordinal field. Tableau has been sorted using a custom sort order (Fatal > Serious Injury > Minor Injury > Damage Only) to have the levels of severity displayed in a sensible, readable hierarchy in all the charts. There was also a uniform color scheme: All/aggregate metrics are blue (#1F4E79), Fatal are red (#C00000), Serious Injury is amber/yellow, Minor Injury is teal (#76B7B2), and Damage Only is green. This color grammar is in line with the existing traffic safety standards and its support of pre-attentive encoding, which enables users to immediately differentiate the levels of severity among charts.
- **Missing Values:** A complete audit showed that 0 null values were created in all 17 attributes which is compatible with the synthetic generation process. The NumberofCasualty field was skewed to the right (mean of 5.08, maximum of 25) with the outliers of high casualties being aggregated in visual summaries.

2.4 Data Suitability

Although it is not real-life, the dataset is very similar to the attributes composing the actual record of road accidents operated by agencies and organizations, including the Department of Transport Management (DoTM) of Nepal and international organizations like the WHO Global Status Report on Road Safety (WHO, 2023). It holds the minimum dimensionality necessary in both time and space, categories and numbers to allow all the six necessary types of visualizations. Its volume of 1,850 records is adequate to detect statistically significant distributions as well as be computationally feasible in Tableau. The known limitation of synthetic origin is acknowledged in Task 5.

Task 3: Visualization Design & Justification

The dashboard brings the eight types of visualizations, each of which is chosen depending on the analytical task, types of data used, and principles of visual encoding (Munzner, 2014; Cleveland and McGill, 1984).

3.1 KPI Metric Cards

On the top of the dashboard, there are six summary tiles (Total Accidents, Total Casualties, Fatal, Serious Injury, Minor Injury, Damage Only) with current-year values and the arrows of the yearly changes of the values. Rationalization: KPI cards take advantage of pre-attentive attribute processing and allow executive stakeholders to scan headline measures within less than two seconds (Few, 2006). Placing them initially is in line with the F-pattern reading principle, the storytelling point of entry to users, prior to usability of in-depth charts. The YoY deltas convert standalone figures into comparative intelligence which is solving Decision Objective number 5.

Figure 1: KPI Metric Cards

There is a second KPI tier in the dashboard that displays the number of casualties by vehicle type Bus (1,643 ▲2.43% YoY), Car (508 ▼2.68% YoY), Motorbike (614 ▲1.82% YoY), and Truck (184 ▲42.64% YoY) with each vehicle being easily identified as such with a vehicle icon. These tiles are entirely reactive to the parameter of Select Accident Severity and Select Province, i.e. when a user selects the Fatal severity or a particular province, all the four values are recalculated immediately. This turns the static summary numbers into a dynamic cross-segment tool with the most alarming result being the 42.64% YoY increase in Truck casualties which has indicated that the vehicle safety emergency is massive and needs to be responded to by the regulatory authority at once.

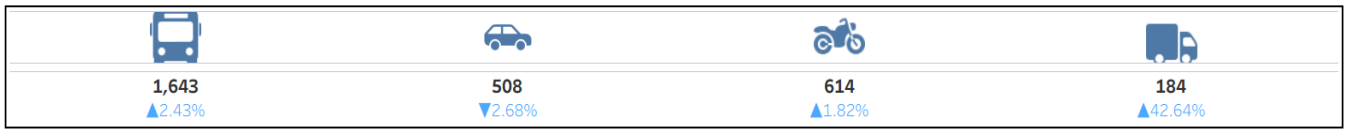

Figure 2: KPI Metric Card for Vehicle Type

3.2 Donut Charts – Weather Condition & Road Surface

There are two donut charts illustrating proportional distribution of casualties by Weather Condition (Clear 42.96%, Rain 31.33%, Fog 8.68, Cloudy 10.31), and Road Surface (Wet 46.69% Slippery 20.07, Dry 22.24, Damaged 10.99) with the total number of casualties (2,949) at the centre. Rationales: Donut charts are suitable when the relationships are part to whole and the number of categories is five or less where the angular comparison is a perceptual truth (Ware, 2012). The donut hole decreases the visual weight in the centre focusing attention to proportional parts. Dominating Clear weather (42.96%) reveals the counterintuitive finding that driver overconfidence, rather than unfavorable conditions might be the major risk factor.

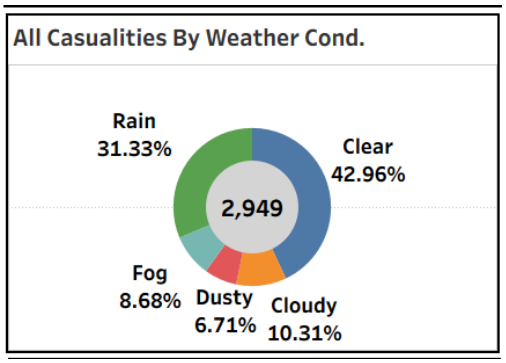

Figure 3 Donut Chart- Type Causalities By Weather Condition

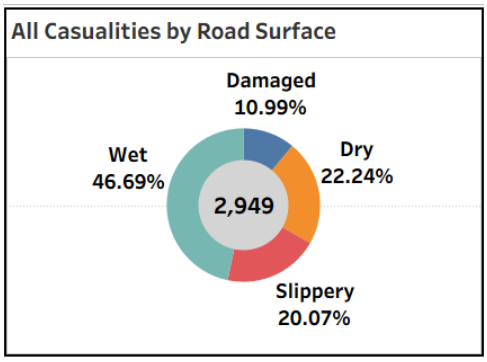

Figure 4: Donut Chart- Type Casualities By Road Surface

3.3 Horizontal Bar Chart – Casualties by Speed Zone

Preference of ranks in terms of the number of casualties: 30 km/h (954) 50 km/h (754) 80 km/h (571). Rationale: Horizontal bars are the best to compare discrete ordered categories that do not have to be rotated with text labels (Munzner, 2014). Ranking by speed zone over casualty count provides an insight into the policy relevant finding in a non-monotonic relationship between speed limit and casualty which indicates low-speed urban areas cause the greatest casualties because of the increased traffic density.

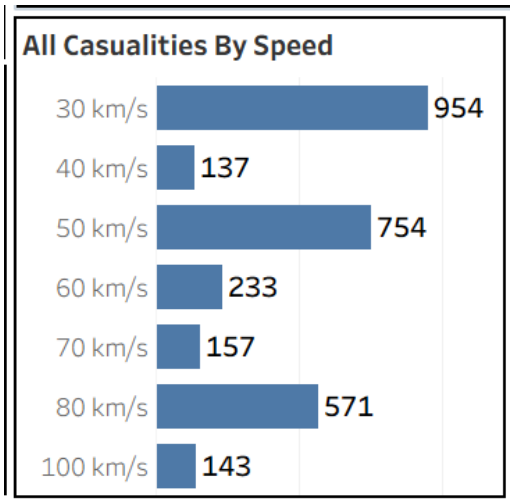

Figure 5: Horizontal Bar Chart - Type Causalities By Speed Zone

3.4 Dual-Layer Area/Line Chart – Monthly CY vs PY Trend

A stipulated area shows the PY fatal casualties per month; an orange line the CY fatal casualty per month. Created by the Current Year and Previous Year parameter. Reason: Trends over time are most effectively represented by a position on the same scale with time being the x-axis (Cleveland and McGill, 1984). The dual-layer (line over area) design does not have the balancing line ambiguity but lets the lines be separated visually. Immediately salient are deviations of the CY line of the PY reference area. Only two years can be compared, and this is in direct response to Decision Objectives 2 and 5, which are dynamic parameters.

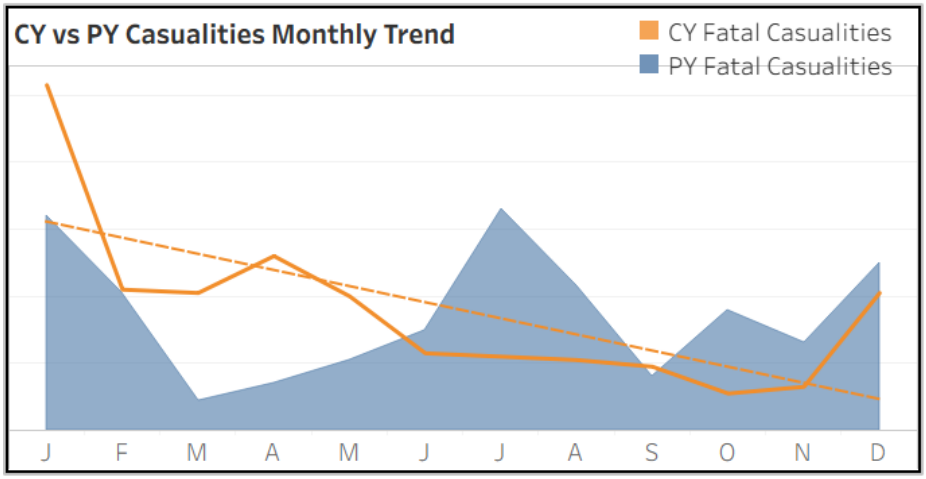

Figure 6: Area/Line Chart- CY vs PY causalities Monthly Trend

3.5 Horizontal Bar Chart – Casualties by Time of Day

Rank 5 periods by casualties: Reason Justification Bar charts employ the length of a bar along a common base to indicate quantity, which is the best perceptual encoding of numerical comparison (Cleveland and

McGill, 1984). It is a very crucial fact that the night shift has the highest percentage of casualties (36.9) and the least traffic as well, which is highly valuable in the scheduling of the patrol so that the highest number of casualties during night could be covered. The severity filter can be used to check whether this concentration at night holds also in case of Fatal records only.

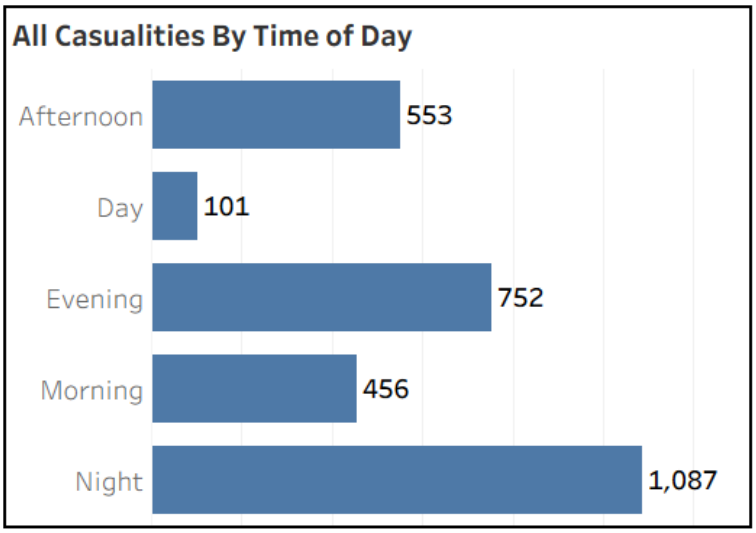

Figure 7: Horizontal Bar Chart - Type Causalities By Time of Day

3.6 Column Bar Chart – Casualties by District

It is a vertical bar chart of casualties in 27 districts in descending order. Surkhet, Kathmandu, Rupandehi and Kapilvastu have the highest number of about 350-420 casualties. Reason: The use of a bar chart over choropleth map in this case is based on the fact that certain comparison of rank of 27 districts is better decoded in terms of bar length than in terms of geographic area fill (Ware, 2012). The provincial filter is doing its job by allowing the ranking to be done on proper administrative boundaries.

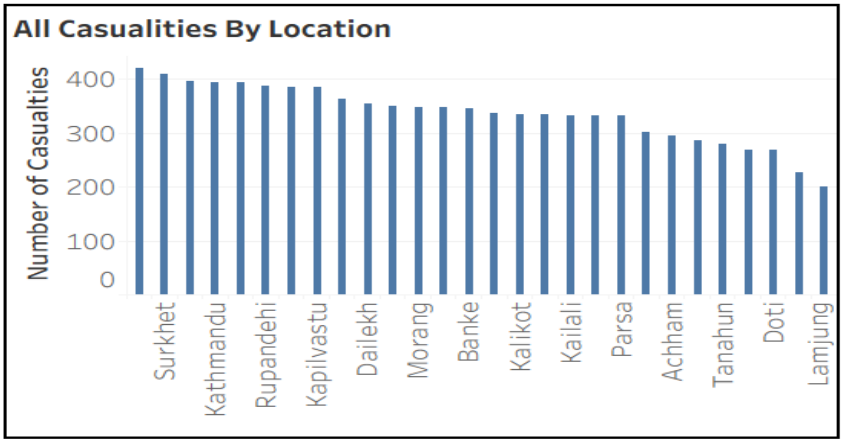

Figure 8: Column Bar Chart- Type Causalities by Location/District

3.7 Grouped Column Chart – Province vs Accident Severity

Comparisons Current Year Casualties show labels of percentages change share between seven provinces: Madhesh (15.67%), Lumbini (14.95%), Koshi (13.73%). Rationale: Grouped bars can be used to make simultaneous comparisons of two categorical variables (province and severity). Cognitive load of reading the axis is eliminated by percentage labels on the top of the bars. The Select Province parameter is able to make drill-down to any given province, which is able to directly aid in decisions relating to inter-provincial resource allocation.

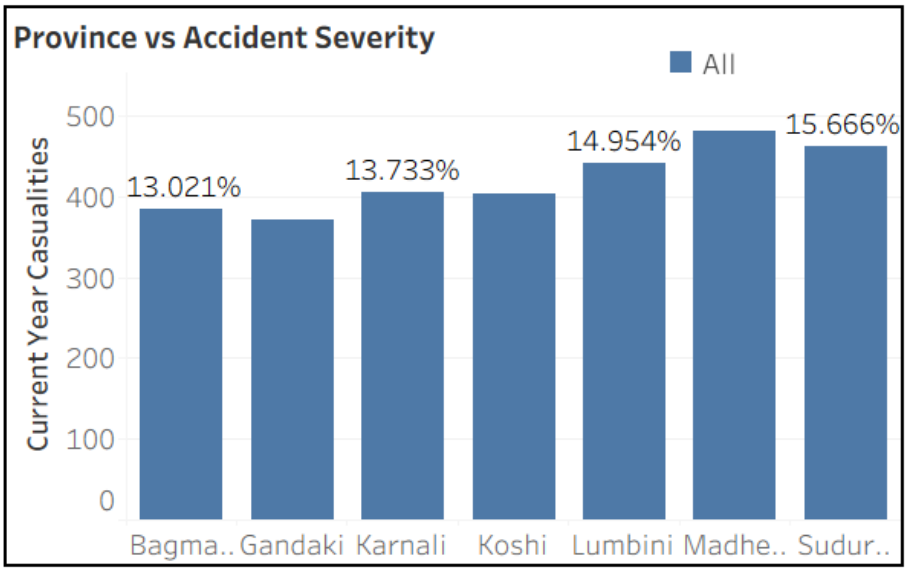

Figure 9: Grouped Column Chart- Province vs Accident Severity

3.8 Horizontal Bar Chart – Casualties by Road Type

Displayed by type of road: Highway (962, 32.6%), Rural (819), Urban (635), Feeder (533. Reasoning: Road type is a nominal variable; bar length indicates magnitude differences quite well. Thee observation that the Highways (even though lower in numbers compared to urban roads) cause the greatest number of the casualties suggests the existence of a high caste per-road severity rate. This, together with the road surface donut, allows the analysis of correlation between the use of highways and wet/slippery roads.

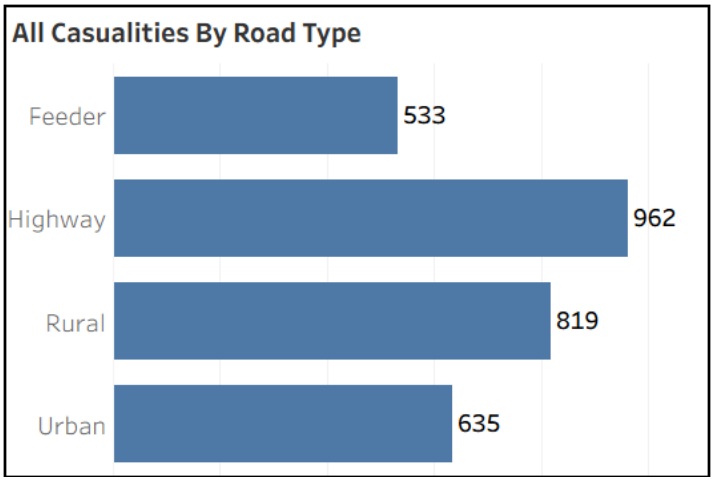

Figure 10: Horizontal Bar Chart- Type Causalities By Road Type

3.9 Donut Chart – Accidents by Driver Gender

Shows accidents based on gender of drivers: Male (381, 64.8%), Female (137, 23.3%), Other (70, 11.9%). Justification: A donut chart is useful with three categories to enable the correct part-to-whole perception. The prevalence of male drivers complies with the international trends of increased rates of male driving and risk-taking behavior in Nepal (WHO, 2023), which is the basis of designing campaigns.

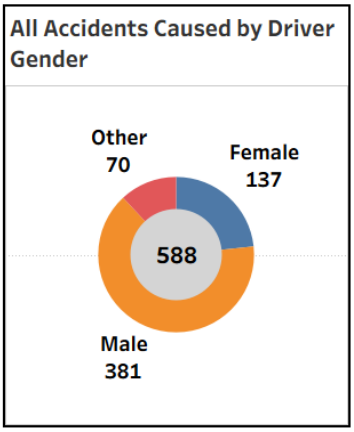

Figure 11: Donut Chart- Accidents By Driver Gender

Task 4: Interactive Dashboard Development

4.1 Dashboard Overview

Tableau was used to build the dashboard in Tableau Desktop and the workbook was exported as twbx. It can support a fixed layout at 1920x1080 resolution display and is spatial consistency across platforms. The title banner has the Nepal flag emoji and the name of the dashboard and the four interactive parameter controls in the upper-right corner, creating an immediate brand recognition and the ability to control the user.

4.2 Interactivity Features

- Current Year / Previous Year parameters can update all CY/PY calculated fields simultaneously, and also allow very easy compare all charts for different years at the same time without getting out of the page.
- Select Accident Severity filter is used worldwide with the formula: [Select Accident Severity] = [Accident Severity] OR [Select Accident Severity] = 'All' to filter a particular severity level the entire dashboard.
- Select Province filter is selected with the use of the Correct Province calculated field: [Correct Province] = [Select Province] OR [Select Province] = 'All' which is spatially accurate through the corrected district mapping.
- All charts have custom tooltips showing precise counts of casualties, severity breakdowns, share of percentages on hover, which is detailed and without visual clutter.

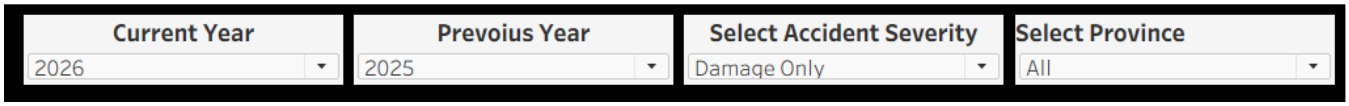

Figure 12: Dashboard Filters Screenshot

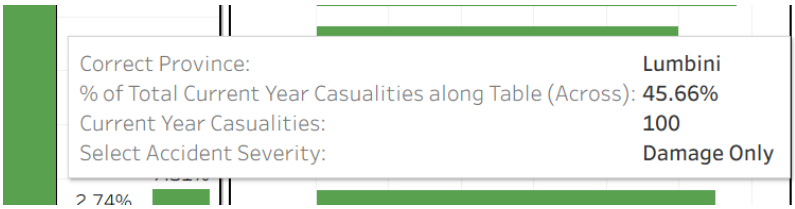

Figure 13: Custom Tool Tips Screenshot

4.3 Layout & Storytelling Flow

The dashboard is designed with a planned narrative that moves from top to bottom, overview to insight, as follows:

- **Level 1 (Top) Executive Summary:** KPI tiles and YoY indicators to have instant situational awareness.
- **Level 2 (Upper Middle) Vehicle Breakdown:** Four types of vehicle metrics that have YoY change put the transport segment trends into context.

- **Level 3 (Middle) Environmental and Temporal Analysis:** Speed, monthly trend and time-of-day charts respond to the question when and under which conditions accidents take place.
- **Level 4 (Bottom) - Geographic and Demographic Detail:** The charts and maps of districts, province-severity, road type, and gender assist in making specific interventions.

This hierarchy is based on the concept of dashboards leading with context and then provide details only (Few, 2006), catering to the needs of the executive as well as the operational analysts.

4.4 Visual Design

Dashboard color scheme is consistent and based on blue, with the accentues saved to the severity coding. Maximum ratio of data-ink is in white backgrounds (Tufte, 2001). The font size is adjusted to ensure that it can be read at 1080p resolution. The regularity with which the Nepal flag icon has been used and the national context used in the title takes the audience in the geographic domain. All charts have grid lines minimized to decrease visual noise and the title of the chart labels descriptive, starting with the word All to indicate that the metric measures all data except when filtered.

Task 5: Insight Generation & Reflection

5.1 Key Insights

Insight 1. the most dangerous time is a night: Night comprises 1,087 casualties, which is almost twice the amount of the afternoon (553) and almost three times more than the morning (456). Although there is less traffic, less visibility, driver fatigue, and speed on off-peak hours, the hazards are disproportional. It has a direct effect on the patrol schedules and investment in street lighting.

Insight 2 Clear Weather brings the Greatest number of casualties: 42.96% of all casualties happen in the clear weather, the only largest category of weather. This is an unexpected result that aligns with findings in the rest of the world (WHO, 2023), indicating that driver overconfidence under favorable conditions is a more significant risk factor than unfavorable weather. It also opposes the enforcement based on weather conditions only and suggests checking all year round.

Insight 3 Wet and Slippery Roads Contribute Two-Thirds of the Casualty: 46.69 percent of the casualty is on wet roads and another 20.07 percent on the slippery roads. This, together with the monsoon season of Nepal (June-September), highlights the need to develop better road drainage systems and enforcement of seasonal tyre control.

Insight 4 Highways Are Disproportionately Dangerous: There are 962 casualties (32.6 percent) on highways (the most of all road types), even though urban roads are more numerous. A high-speed highway risk profile (80 km/h) with the second-largest number of casualties (571) supports this claim.

Insight 5 Madhesh and Lumbini Are the Two Worst-Off Provinces: It has 15.67 percent and 14.95 percent of total casualties and should be the focus of road safety investment and heavy vehicle control since it has many highways in place and a large volume of commercial traffic.

Insight 6 Over-representation of Male Drivers: Male drivers over-represent in the accident records with 64.8% of them. Together with the fact that motorbikes are the most accident-prone vehicle category (520 records), specific safety messaging among young male motorcyclists forms one of the areas with the highest impact of an intervention.

5.2 How Visualization Answers the Problem

The interactive dashboard communicates with all the five decision-making objectives. Geographic concentration can be viewed in the district and province charts, temporal in the time-of-day and monthly

trend charts, environmental risk factors with the weather and road surface donuts, vehicle and demographic with the vehicle breakdown and gender donut, and year on year trends with the YoY KPI tiles and the comparative trend line. The interactivity based on parameters indicates that a traffic commander in Madhesh can select to their province and the Fatal severity to specify the particular districts, road types, and the time periods that are most essential to them in their jurisdiction.

5.3 Limitations

- **Synthetic data origin:** The data was created artificially and not intended to be used directly in actual policy formulation without first being checked against real DoTM records.
- **No geographic coordinates:** GPS data is not provided, so it is impossible to map the road segments on a spatial level and consider it only at the level of districts and provinces.
- **Correlation, not causation:** The dashboard shows the associations, but not the causal relationships. The confounding factors like traffic density, experience of drivers and the level of enforcement are not present.
- **Incomplete 2026 data:** Since there are only 107 records in January of 2026, any comparison of trends with 2026 as the current year would give bias in the monthly chart.

5.4 Future Improvements

- **Actual DoTM data integration:** The substitution of artificial data with the confirmed records would turn it into a real operation policy instrument.
- **Geospatial map layer:** With choropleth map (based on district-level coordinates), the non-technical stakeholders would be able to have intuitive spatial visualizations.
- **Predictive analytics:** The inclusion of in-built forecasting (exponential smoothing) offered by Tableau would forecast the trends of the number of casualties within the next 12 months, facilitating upstream policy actions.
- **Driver age dimension**: Inclusion of age data would determine high risk groups, and this would be important given that Nepal has many young motorcyclists.

Dashboard Screenshot

Figure 14 demonstrates the default view of Nepal Road Safety Analytics Dashboard (2023-2026) in the main Tableau dashboard (Current Year: 2026, Previous Year: 2025, All Severity, All Provinces). **The dashboard is also accessible via Tableau Public:**

https://public.tableau.com/shared/9GRWQD7BC?:display_count=n&:origin=viz_share_link

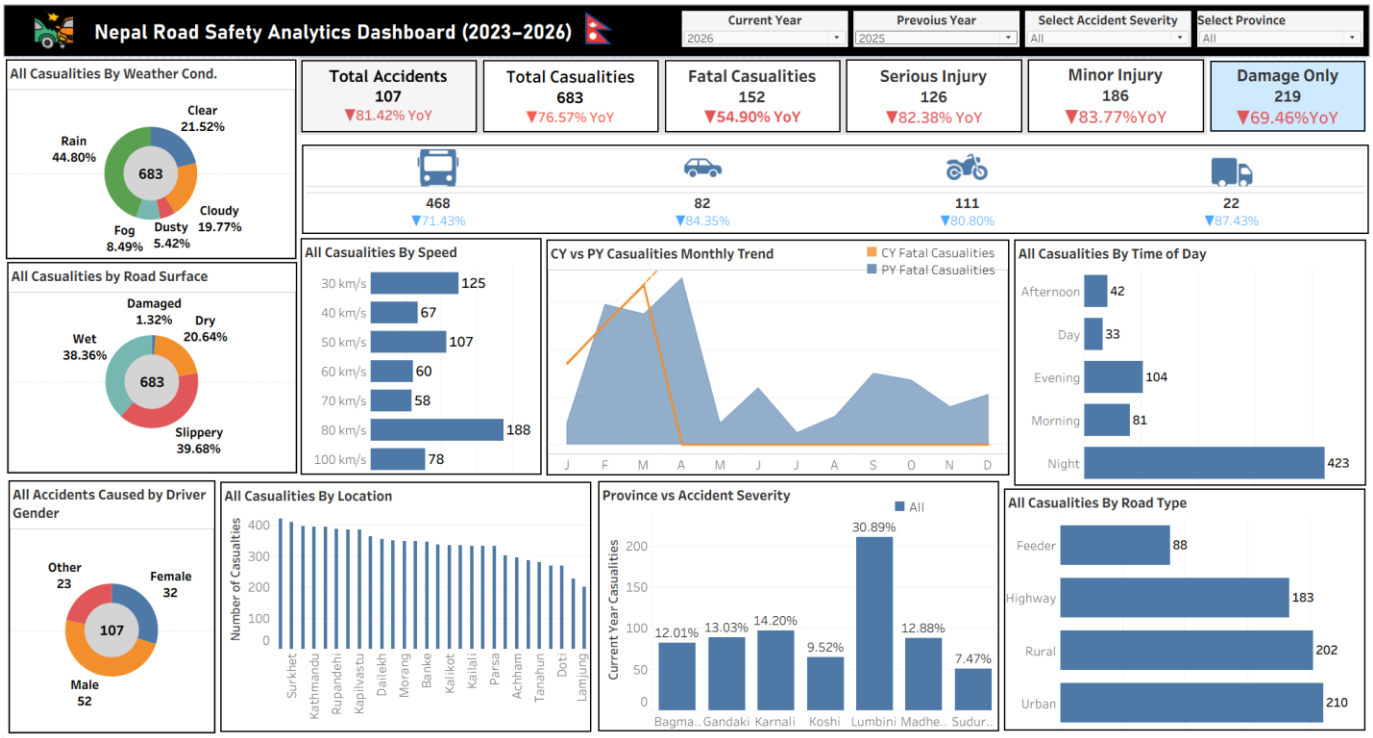

Figure 14: Nepal Road Safety Analytics Dashboard(2023-2026) Screenshot

Figure 14: Nepal Road Safety Analytics Dashboard (2023–2026) – Tableau Desktop

The dashboard is presented in the state of the filter (Current Year: 2026, Previous Year: 2025, All Severity, All Provinces), where the total accidents represent the number of accidents in the current year and are 107, the total number of accidents is 107, and the total number of casualties is 683. The main metrics that are visible are 152 fatal casualties (-54.90% YoY), night-time peak casualty of 423, highway road type with the highest number of casualties at 183, and the largest share of Lumbini Province with 30.89. The top-right corner clearly shows all four parameter controls (Current Year, Previous Year, Select Accident Severity, Select Province), which proves the entire dashboard to be actively interactive.

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